Assignment KIE4031 (20%)

Emerging technologies in smart cities focus on leveraging IoT, AI, and 5G to enhance urban efficiency, sustainability, and quality of life. Key innovations include digital twins for urban planning, AI-driven traffic management, smart energy grids, automated waste management, intelligent street lighting and more. These technologies transform data into actionable insights for safer, cleaner, and more responsive cities.

Assignment tasks :

1) Identify real-world problems related to smart cities technologies. Perform case study on ONE innovative technology in smart city. Describe the features and innovation.
2) Investigate the incorporation of machine learning models (such as regression, classification and clustering approaches) in the chosen technology.
3) Propose a framework of ONE innovative technology in smart cities. Then, by using public dataset, develop a machine learning based model to demonstrate the proposed framework in solving problems and challenges in smart cities technologies.
4) Discuss future potential of the chosen machine learning model and ethical issues such as: data privacy, bias in algorithms, security concerns.

Submission :

1) This is a group assignment. Each group can have 2-3 students. Each group will need to submit 1 report and 1 presentation only. Please submit your assignment via SPECTRUM before the deadline.

2) Prepare a report (maximum 15 pages with font size '12' and single spacing). Please include the link to the codes used in this assignment.

3) Your scientific report should include [C6]:
    i. Background and case study
    ii. Methodology and flow diagram of the proposed model
    iii. Results and discussion
    iv. Future works
    v. Ethical issues

4) Prepare presentation slides (maximum 5 slides) and present in class.
    i. Work development [P2]
        - Explain the workflow diagram of the proposed machine learning model
        - Present the output of the proposed machine learning model
    ii. Response and reflection [A2]
        - Discuss on technology impact and your perspectives on the emerging technology
        - How do you feel about technological change?